

Coding Theory: The McEliece Public-Key Cryptosystem

Bernd Stelzl, Sebastian David Albu, Cryptanalysis 2024

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Coding Theory

- Mathematical field...
 - ...on transmitting messages over **symmetric channels**.
 - ...while accounting for transmission **errors**
 - ...concerned with finding such **ideal codes** to optimize **error-detection** and **correction**
- **Cryptographic coding** emerged later as subfield
- We are mostly interested in **linear codes** which give structure

Mathematical Preliminaries

Finite Field $\mathbb{F}_q$

Definition (Field)

A set F with functions $+: F \times F \rightarrow F$ and $\cdot: F \times F \rightarrow F$ is called a **field**, if

- $(F, +)$ is an **abelian group**
- $(F \setminus \{0\}, \cdot)$ is an **abelian group** and
- the **distributive** property holds for $\{+, \cdot\}$

Finite Field $\mathbb{F}_q$

Definition (Finite Field)

A **finite field** is a field which contains finitely many elements. We write $\mathbb{F}_q$ to denote a **finite field** with q elements.

- We are interested in prime fields
 - Constructed as $x \bmod p \Rightarrow \mathbb{F}_p$
 - In practice, $p = 2$ (binary field)

Vector Space $\mathbb{F}_q^n$

- No need to fully recall the definition of vector spaces...
 - ...just remember it gives us nice **structure**
- $\mathbb{F}_q^n$: n -dimensional vector space over $\mathbb{F}_q$.
- Linear codes are linear subspaces of $\mathbb{F}_q^n$.
 - Think: Bit-vectors
 - This motivates the idea of **distance**

Linear Codes

- $[n, k, d]$ -code C :
 - n : length of codewords
 - k : dimension of the code
 - d : minimum distance between codewords
- As linear subspace, a linear code gives us...
 - Additive Properties $\forall \mathbf{u}, \mathbf{v} \in C : \mathbf{u} + \mathbf{v} \in C$
 - Scalar Multiplication

Codewords and Generator Matrix

- Codewords: n -dimensional vectors in C .
- Generator matrix G : $k \times n$ matrix.
 - G then generates every codeword in C
 - $\mathbf{c} = mG \in C$ for $m \in \mathbb{F}_q^k$.

Goal: Error-correcting Codes

- Codeword $\mathbf{c}$ sent over an **symmetric channel**.
 - Where each symbol has an **error probability** p
- Error vector $\mathbf{e}$: $\mathbf{y} = \mathbf{c} + \mathbf{e}$.
- Detect erroneous codewords received
- Recover original codeword $\mathbf{c}$ if errors are small.

Hamming Distance

- Hamming distance $d_H(\mathbf{x}, \mathbf{y})$: calculated using Hamming-Weight w_H .
- Intuitively: Number of differing bits
- Expressed in terms of **Hamming-Weight**:
 - $d_H(\mathbf{x}, \mathbf{y}) = w_H(\mathbf{x} - \mathbf{y})$.

Minimal Distance

- Minimal distance d : Minimum Hamming-Weight of non-zero codewords in C .
- Error-correcting capability $t = \lfloor \frac{d-1}{2} \rfloor$.
- Intuition
 - Codewords as vectors
 - Correction possible when correct word closest
 - Detection possible when received word is not a codeword

Dual Codes

- Dual code $C^\perp$: vectors orthogonal to C .
- $C^\perp = \{\mathbf{w} \in V(n, q) \mid \mathbf{u} \cdot \mathbf{w} = 0 \ \forall \ \mathbf{u} \in C\}$.
- Parity check matrix H : generator matrix of $C^\perp$.
- Parity check matrix H : used for error correction

Syndrome Decoding

- Parity check matrix H of an $[n, k]$ -code C .
- Syndrome $S(\mathbf{y}) = \mathbf{y} \cdot H^T$.
- $S(\mathbf{y}) = 0$ iff $\mathbf{y} \in C$.
 - $\Rightarrow$ Otherwise an error is present.
 - Precompute common error vectors, compare, and correct.

The McEliece Public-Key Cryptosystem

Setup

- Based on hardness of decoding.

Public key:

- $G' = SGP$
- t : max number of correctable errors

Secret keys:

- S : random invertible $k \times k$ matrix
- G : generator matrix of the original code
- P : random $n \times n$ permutation matrix

Encryption

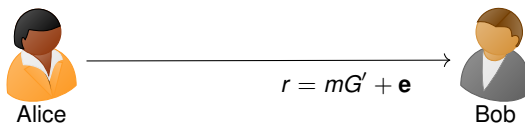
- Alice sends message m encrypted:

$$r = mG' + \mathbf{e},$$

with $G' = SGP$ (public key).

- G' appears random, making decoding hard
- Error vector $\mathbf{e}$ with $w(\mathbf{e}) \leq t$.

Encryption



Decryption

- Bob computes: $rP^{-1} = (mS)G + \mathbf{e}P^{-1}$.
- Private parameters: S, G, P . Only G' known.
- Uses decoding algorithm to find mS .
- Recovers m using S .

Parameters and Security

- Original scheme: binary codes, $n = 1024$, $k \approx 524$, $t = 50$.
 - n : length of codewords
 - k : dimension of the code
 - t : max number of correctable errors
- Parameters chosen to make decoding infeasible.

Modern Considerations

- Original parameters outdated.
- Still considered post-quantum secure. (Larger parameters)
- No widespread real-life applications as of 2016.

Summary: Coding Theory and McEliece Introduction

- Coding theory uses linear algebra and finite fields.
- Linear codes and dual codes are fundamental.
- Error correction and McEliece cryptosystem are key applications.

Adaptive Chosen Ciphertext Attack

Adaptive Chosen Ciphertext Attack

- Proven existence of an adaptive chosen ciphertext attack on the McEliece cryptosystem (Trevisan, 2004).
- The attack can be summarized in the following steps.

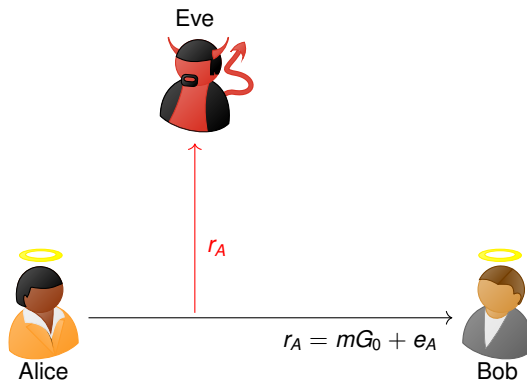
Maximal Error Property

- On input of a vector $y \in \mathbb{F}_q^n$, the decoding algorithm A_t returns a codeword c in C at distance $\leq t$ to y if such a codeword exists.
- Otherwise, it returns an error message.
- The algorithm A_t will not try to correct $t + 1$ errors.
- If too many transmission errors occur, r may be at distance $\leq t$ from another codeword.
- In this case, A_t will not return an error message.

Setup

- Assume the decoding algorithm A_t used in a McEliece cryptosystem has the maximal error property.
- Let r be the ciphertext Alice sent and Eve intercepted, where $r_A = mG' + e_A$, the A in r_A stands for Alice.
- Bob never receives the original r_A .

Ciphertext Interception

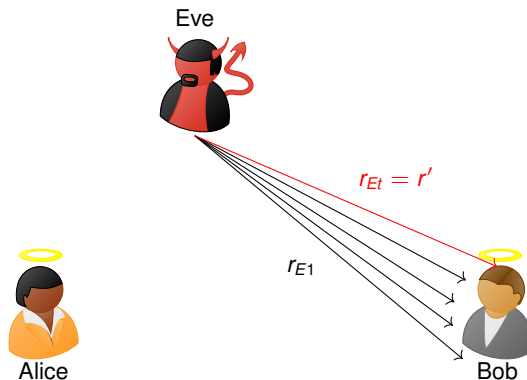


Step 1: Increase the Number of Errors to Exactly t

- Eve repeatedly changes a random coordinate in r_A and sends the altered codeword r_E to Bob.
- Continues until Bob returns an error message indicating one error too many.
- The previous message contains exactly t errors.
- Eve proceeds with this message r' to the next step.

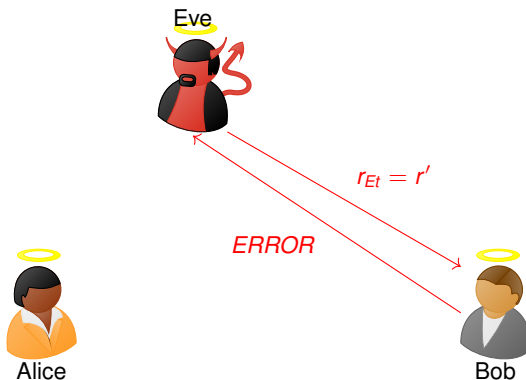
Adaptive Chosen Ciphertext Attack on McEliece

Sending until Bob returns Error



Adaptive Chosen Ciphertext Attack on McEliece

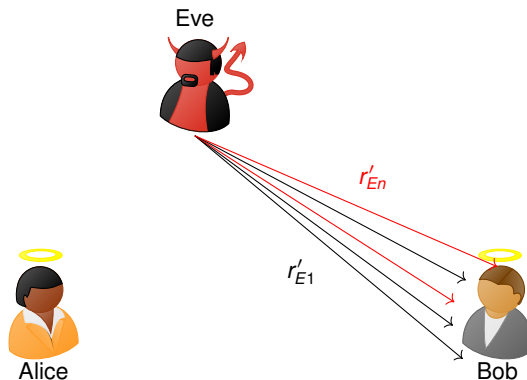
Sending until Bob returns Error



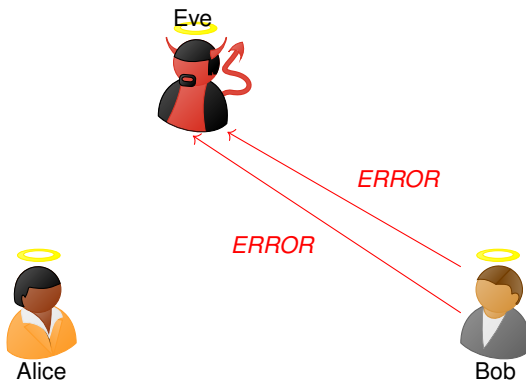
Step 2: Determine Error-Free Coordinates

- With r' containing exactly t errors, Eve changes random coordinates in r' and sends these to Bob.
- Identifies error-free coordinates when Bob returns an error message.
- Continues until enough error-free coordinates are identified for the next step.

Determine Error-Free Coordinates



Determine Error-Free Coordinates



Step 3: Determine the Plaintext

- Solves the matrix equation $r' = mG' + e'$ for the plaintext m .
- Crucial step to isolate m from r' .
- Uses Gaussian elimination on the columns corresponding to the known error-free coordinates.
- Gaussian elimination reduces the system to a simpler form, making it possible to solve for m efficiently.

Conclusion of the Attack

- With the above steps, Eve completes an attack using adaptive chosen ciphertexts.
- In theory, this attack is proven correct by Trevisan (2004).
- In practice, no real applied attack has been conducted due to the absence of the McEliece scheme in real-world applications.

Countermeasures

- Trevisan (2004) proposed several countermeasures to counter adaptive chosen ciphertext attacks (CCA) on the McEliece cryptosystem:

Countermeasure 1: Checking for Repeated Messages

- Bob can check for repeated messages to detect attacks.
- However, Eve can bypass this by adding a random codeword from the code C to each probe, maintaining the same error vector e .

Countermeasure 2: Fixed Error Vector Weight

- Fix the error vector's weight to t (or $t' \leq t$).
- If an error vector of different weight is encountered, return an error message.
- Effective attacks remain possible for non-binary and binary codes by probing coordinates.

Cryptanalysis of McEliece

Introduction

- Attempts at constructing attacks have been many after first proposal
- Based on the publication by Korzhik and Turkin (1991).
- Due to the underlying complexity, attacks are not easy to understand
 - We want to give some intuition on where we are, and what to look out for

First Approach: Finding Nonsingular Submatrices

- Recall: Non-singular $\Leftrightarrow$ Invertible $\Leftrightarrow$ Determinant $\neq 0$.
- Find k coordinates where the matrix G' contains a nonsingular $k \times k$ submatrix.
- Ensure the error vector $\mathbf{e}$ has zeros in these positions.
- Makes decoding feasible by inverting the submatrix.

Second Approach: Equivalent Generator Matrices

- Leverage the fact that G' might be equivalent to the generator matrix of another Goppa code.
 - Finding such G'' possibly reveals structure.
 - In some instances allows the attacker to correct the error, bypassing original Goppa-code.
- The probability of this occurring is very small.

Third Approach: Iterative Optimization Algorithm

- Uses an iterative optimization algorithm.
- Guarantees correction of error patterns with weight up to $\lfloor (d - 1)/2 \rfloor$ for any linear code with minimum distance d .
- Choice of optimization algorithm may depend on code.
- Computational power limits feasibility.

Iterative Optimization Algorithm: Process

1. Embed binary vectors into the continuous vector space $\mathbb{R}^n$.
 2. Reduce the decoding problem to finding extrema in $\mathbb{R}^n$ under code constraints.
 3. Use an iterative procedure to find the desired extremum for error patterns with weight at most $\lfloor (d-1)/2 \rfloor$.
- This approach does not contradict the NP-completeness of the general decoding problem.
 - Applies only to words within a specific distance from a code word.

Application to (63, 26) Code

- Applied to a (63, 36) BCH code with minimum distance $d = 11$.
- Transformed the generator matrix G using random nonsingular and permutation matrices to get G' .
- Tested cryptanalysis with error patterns of Hamming weight 5.

Results of Application

- In 100 trials, the correct message was found every time.
- Required about 10^6 operations, significantly less than the 5×10^6 operations for an exhaustive attack.

General Application

- For an arbitrary (n, k) code, this attack requires approximately $2^{0.113n}$ operations.
- For a $(1024, 654)$ BCH code with $d = 65$, the attack was demonstrated on an IBM PC during Eurocrypt '91.
- Back then, it took about 60 hours.

Post-Quantum Security of McEliece

Introduction

- The cryptosystem carries relevancy especially due to longevity
- A finalist in the NIST PQC standardization process.
- But despite strong security claims, poses practical challenges

Key Size Challenge

- Large key size poses a problem to many areas of applications.
 - $1\text{MB} \approx 256\text{-Bit Security}$
- Approaches to key size reduction were proposed over the years...
 - But eventually led to algebraic attacks breaking security claims
- Some new ideas were proposed (2021), but time will show...

Security Properties

- The basic variant of the McEliece scheme offers promising properties in PQC.
- No polynomial-time quantum algorithm has been found so far.
- The incorporation of randomness could lead to further applications in privacy related schemes.

Challenges of Code-based Schemes

- Few coding-theory schemes aside from McEliece.
- The Niederreiter algorithm is a promising scheme.
- Raises questions compared to other cryptographic problems with many implementations.
 - Long-standing vs. Lack of surface for cryptanalysis

Comparison with Lattice-Based Schemes

- Coding-theory schemes show similarities to lattice-based schemes in many aspects.
- Allows for partial security analysis through reductions between the two mathematical objects, as seen by Wang et al.

Need for a Larger Selection of Codes

- A larger selection of codes for building cryptosystems would create more resilience.
- Important if efficient quantum algorithms on specific sets of codes are found.
- $\Rightarrow$ This is the core research area in coding theory.

Further Courses

Coding and Cryptography

- Codierung und Kryptographie by Prof. Elsholtz
 - In the current curriculum for Information Security (D3)
 - Offered every two years at TU Graz as seen [here](#)
 - Some old exercise sheets are published [here](#)
- Or if you want a full-text source:
 - R. Hill, "A first course in Coding Theory"

References

- MacWilliams and Sloane, "Theory of Error-Correcting Codes"
- Berlekamp, McEliece, and van Tilborg, "Inherent Intractability of Certain Coding Problems"
- McEliece, "A Public-Key Cryptosystem Based on Algebraic Coding Theory"
- Trevisan, "Some Applications of Coding Theory in Computational Complexity"
- Faugère et al., "Structural Cryptanalysis of McEliece Schemes"
- Wang et al. "Quantum-safe cryptography: crossroads of coding theory and cryptography"

Questions?